

Lecture 19: Clinical & Biomedical Text

Data Modalities

BINF 4002 — Machine Learning for Health

Tuesday, March 31, 2026

Columbia University Irving Medical Center

Where Are We?

Lectures 17–18: EHR data is a **byproduct of clinical care and billing**. A patient's record is a timeline of multi-modal observations.

Today: We zoom into the **text** on that timeline — the richest, messiest, and most rapidly changing part of the EHR.

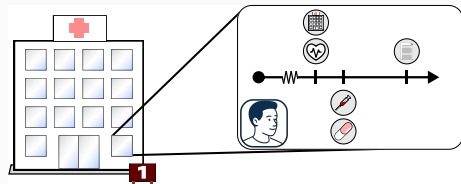
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Running thread: Text is a *sequence of nominal tokens*. We'll see **five ways** clinical text breaks that abstraction. Watch for the **Data type connection** boxes.

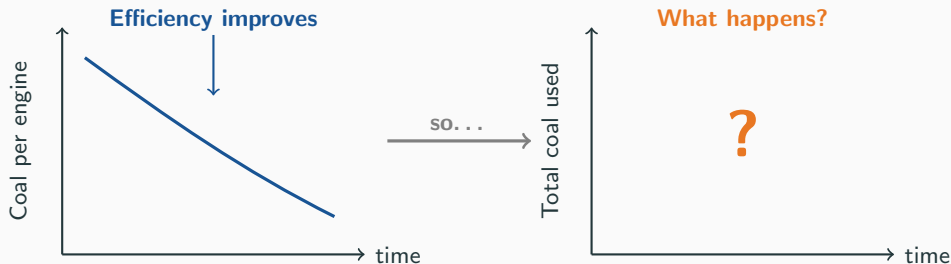
Notebook: `nb19_clinical_text_data.ipynb`



A Paradox to Keep in Mind

A Puzzle from 1865

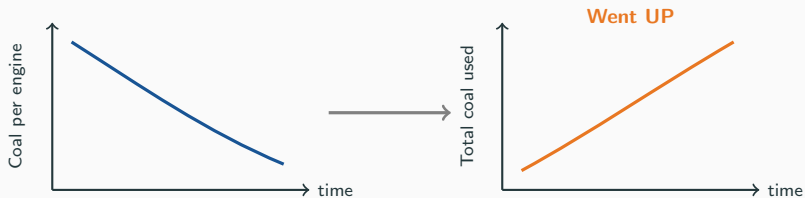
England, 1860s. Steam engines are getting more efficient every decade.



Each engine uses *less* coal. Surely total coal consumption goes *down*?

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The Jevons Paradox: Making something easier to produce often yields *more* of it, not less effort overall. **Keep this in mind** as we tour clinical text.

Who Writes Health Text, For Whom, and Why?

A Taxonomy of Health Text by Communicative Purpose

Health text is not one thing. Four broad categories:

Category	Writer → Reader	Purpose	Examples
Clinical documentation	Clinician → Record	Legal, billing, continuity	Progress notes, H&Ps, discharge sums
Clinical communication	Clinician → Clinician/Patient	Handoff, referral	Rad/path reports, consults, AVS
Patient-generated	Patient → World/Clinician	Help-seeking	Portal messages, forums, social media
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Not exhaustive — also regulatory text, insurance narratives, public health reports. But these four cover most of what ML researchers encounter. We'll take them in order.

Clinical Documentation

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Key framing: Everything that follows — templates, copy-paste, abbreviations — is a **rational response** to writing notes that must serve all three purposes under extreme time pressure.

[Notebook Cell 3: loads MTSamples, ~5K clinical transcriptions across 40 specialties.]

Who Writes Clinical Notes, and Under What Conditions?

Recall from Lecture 17: EHR data reflects what was *documented*, not everything that *happened*.

The clinician's reality:

- US physicians spend **~2 hours on documentation for every 1 hour of patient contact** (Sinsky et al., 2016).

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- “Pajama time”: charting after hours, evenings, weekends.
- Notes must satisfy billing, legal, *and* clinical needs — often in tension.
- Time pressure leads to shortcuts: templates, copy-paste, abbreviations.

Who Generates What? A Cast of Authors

“Clinical notes” are not all written by the same person:

Author	Typical notes	Characteristics
Attending physician	H&Ps, progress notes, discharge summaries	Most autonomy; may dictate or use scribe
Resident / fellow	Same, co-signed by attending	Learning; notes often more verbose
Nurse / RN	Nursing assessments, flow-sheets, triage	Frequent, structured, shift-based
APP (NP/PA)	Progress notes, procedure notes	Similar to physician; scope varies by state
SW, PT, OT, dietitian	Discipline-specific assessments	Different vocabulary and templates

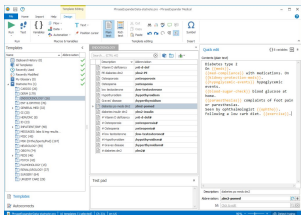
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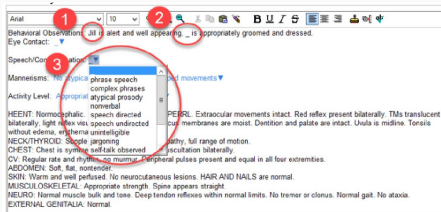
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For ML: Author identity may or may not be in your dataset. Note style, vocabulary, and length vary *systematically* by role — an unobserved confounder if you ignore it.

The Template-Free-Text Hybrid



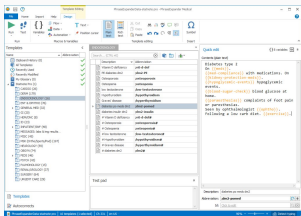
(a) Template expansion: ;dm2-pomed generates a full diabetes note with variables.



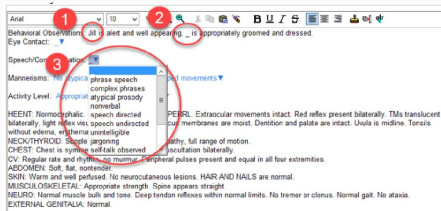
(b) EHR template: dropdowns constrain input to pre-defined options.

Result: Most “text” in a clinical note is **machine-generated boilerplate**. The clinician's actual reasoning is often just 1–2 sentences in the Assessment & Plan.

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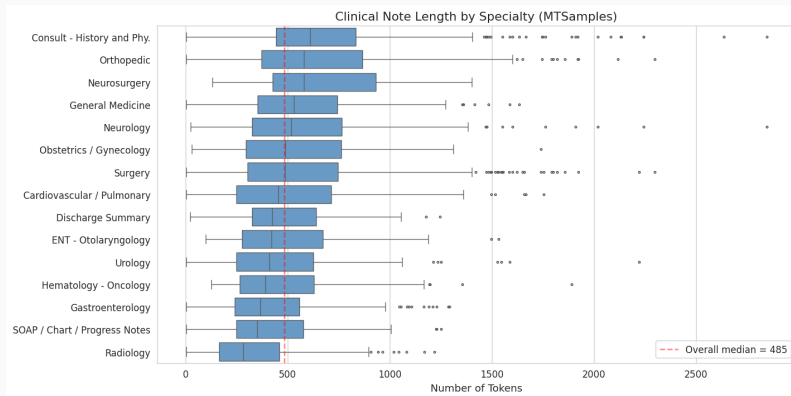


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Result: Most “text” in a clinical note is **machine-generated boilerplate**. The clinician's actual reasoning is often just 1–2 sentences in the Assessment & Plan.

Key implication: An LLM trained on clinical notes is partly learning to emulate **template engines and copy-paste** — generative processes that were already automated *before* any ML entered.

How Much Variation? Note Length by Specialty



Discharge summaries and surgical notes can be 10× longer than radiology reports. “Clinical text” is not one distribution — it’s many. [Notebook Cell 4]

Studies estimate **50–80%** of text in clinical notes is duplicated from prior notes or templates (Tsou et al., 2017; Cohen et al., 2013).

Why this matters for ML:

- **Information content $\neq$ text volume.** A 10-page note may contain 1 page of new information.

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Copy-Paste & Note Bloat

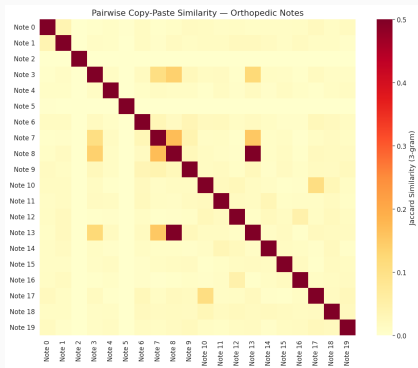
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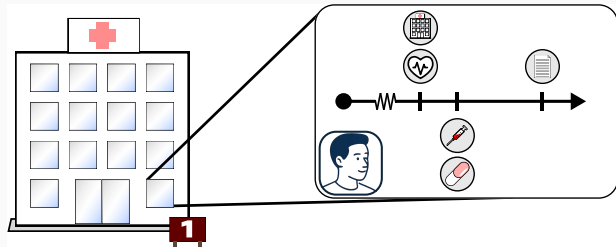
Data type connection: Each note is part of a **temporal sequence** of notes for the same patient. Copy-paste means successive tokens are **not i.i.d.** — 70% duplication shatters the independence assumption.

Copy-Paste Evidence: Between-Patient Template Overlap



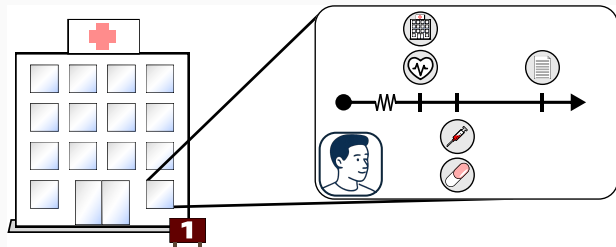
Pairwise Jaccard similarity within one specialty. MTSamples lacks longitudinal patient data, so this is *between-patient template overlap* — real EHR within-patient overlap would be much higher. [Notebook Cell 6]

Text in the Patient Timeline



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Copy-paste across visits is why the i.i.d. assumption fails. The COMPARISON line in radiology reports is an explicit temporal link. And notes reference images and labs that aren't in the text at all — **multi-modal linkage** that a text-only model will miss.

Abbreviations

Abbreviations are pervasive and ambiguous:

Abbrev.	Possible meanings
MS	Multiple sclerosis, mitral stenosis, morphine sulfate, mental status, ...
CP	Chest pain, cerebral palsy, cleft palate
PE	Pulmonary embolism, physical exam, pleural effusion

Disambiguation requires *context*: the same abbreviation means different things in a cardiology note vs. a neurology note vs. a pharmacy order.

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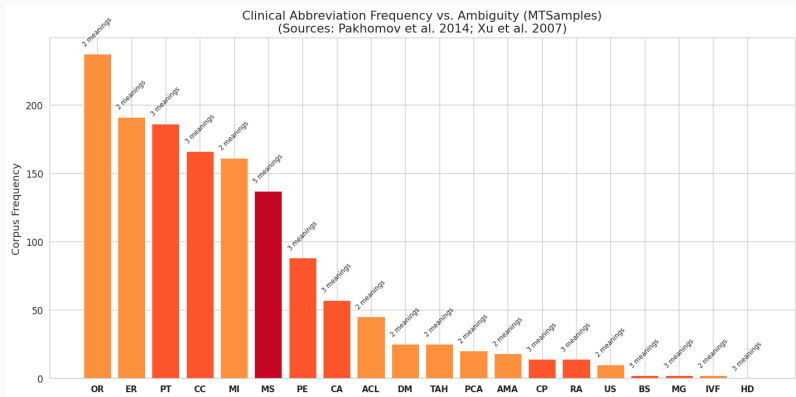
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Data type connection: Tokens are nominal, but their meaning depends on **section structure**. “ROS” vs. “Assessment” headers change what “MS” means — positional context within the note is semantic, not just sequential.

How Ambiguous Are Clinical Abbreviations?



Frequency vs. number of meanings for 20 abbreviations validated against Pakhomov et al. (2014) and Xu et al. (2007). [Notebook Cell 7]

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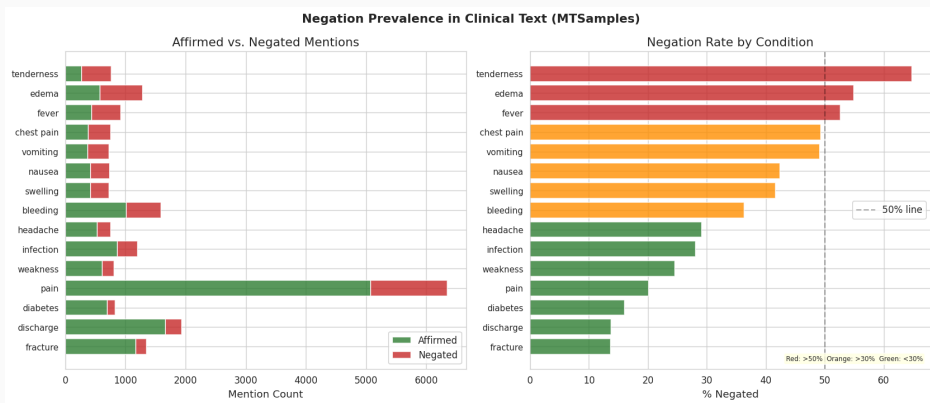
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Dedicated clinical negation tools (NegEx, ConText, NegBio) exist precisely because general-purpose NLP methods underperform at this density.

Note: Negation isn’t unique to clinical text — sentiment analysis (“not bad”) and legal text (“no liability”) face it too. But the *rate* is far higher in clinical notes than in any general-domain corpus.

How Much Negation? Prevalence by Condition



Rash and tenderness are negated >60% of the time — a keyword model would invert their meaning. [Notebook Cell 8]

Clinical Communication

Text Written for Other Clinicians

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Radiology / pathology reports: A specialist interprets a study and communicates findings to the ordering provider.

- Semi-structured: Indication, Technique, Findings, Impression.

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Consult notes, referral letters: One specialist writes to another. More structured than progress notes.

Key difference from documentation: Less copy-paste, more targeted communication. But still templated and billing-driven.

Example: A Radiology Report

INDICATION: 45F, cough, fever, r/o PNA.

TECHNIQUE: PA and lateral CXR.

COMPARISON: Prior 2025-12-01.

FINDINGS: Heart size normal. Lungs clear
bilat. No consolidation, effusion, or PTX.

Osseous structures intact.

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Data type connection: Three data-type issues in one report: (1) section headers change token semantics, (2) COMPARISON line encodes **temporal** dependence, (3) the report **references an image** — text alone is incomplete (**multi-modal linkage**).

Text Written for Patients

A growing category, especially post-21st Century Cures Act (2021), which gave patients access to their own notes (“OpenNotes”).

After-visit summaries (AVS): Auto-generated plain-language summaries given to patients after an encounter.

Patient-facing letters: LLMs are increasingly used to draft replies to portal messages, lab result explanations, post-discharge instructions.

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Privacy & de-identification: HIPAA’s 18 identifiers must be removed before research use. De-id tools replace PHI with tokens like [DATE], [NAME] — these artifacts become part of your data distribution.

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Key challenge: The vocabulary gap. Patients and clinicians describe the same conditions using different words. Models trained on clinical text may fail on patient text, and vice versa.

Scientific & Biomedical Literature

PubMed: 36M+ abstracts. **PMC:** Full-text open-access subset. Fundamentally different from clinical text:

Property	Clinical Notes	PubMed
Purpose	Documentation / billing	Knowledge dissemination
Author	Clinician (time-pressured)	Researcher (edited, reviewed)
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Controlled vocabularies: Papers aren't *written in* controlled vocabularies — they use many of the same terms. **MeSH** is *metadata*: NLM indexers tag articles with MeSH headings *after* publication. **UMLS** links terminologies (ICD, SNOMED, MeSH) into a metathesaurus.

What Does Each Text Type Actually Look Like?

Clinical Note

63M w/ h/o HTN, DM2, CKD3 p/w AMS. Pt found down at home, GCS 11. Labs: Cr 4.2 (baseline 2.1), K 6.1. A&P: AKI on CKD, likely prerenal 2/2 dehydration. IVF NS @ 150cc/hr.

PubMed Abstract

BACKGROUND: Acute kidney injury (AKI) complicates 10--15% of hospital admissions and is associated with increased mortality. **METHODS:** We conducted a retrospective cohort study of 12,413 patients...

Patient Portal Message

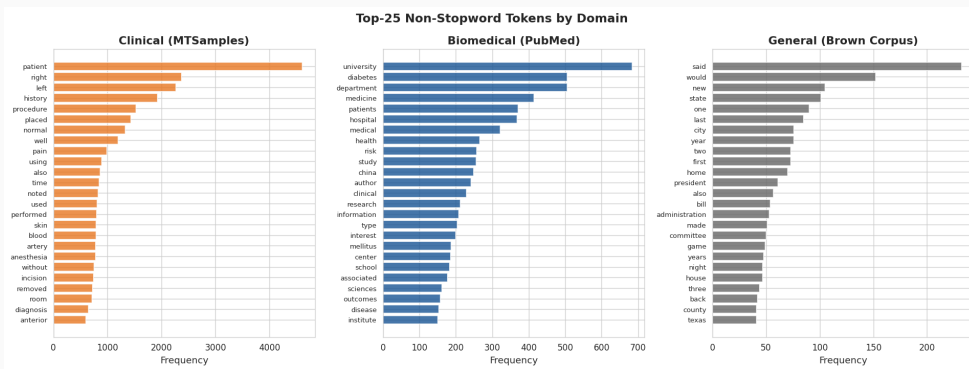
Hi Dr. Smith, I got my blood work back and my creatinine is 4.2 which seems really high?? I've been feeling tired and not eating much. Should I go to the ER or wait until Thursday?

Medical Textbook

Acute kidney injury is defined as an abrupt decrease in renal function, resulting in retention of urea and other nitrogenous waste products. The KDIGO criteria classify AKI into three stages...

Same clinical situation, four texts. Notice abbreviation density, formality, and assumed reader knowledge. [Notebook Cell 10]

The Vocabulary Gap Across Domains



Top-25 non-stopword tokens from clinical notes, PubMed abstracts, and general English. Almost zero overlap — a model trained on one domain will struggle with the others. [Notebook Cell 5]

How Does Biomedical Text Differ from Other Academic Text?

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This is why domain-specific models (PubMedBERT, BioGPT, SciBERT) consistently outperform general models on biomedical benchmarks.

Where do researchers get health text data?

- **MIMIC-III/IV**: De-identified ICU notes, credentialed access. Most-used clinical NLP dataset — but skewed toward critical illness.
- **MTSamples**: Open-access transcribed reports. Small, noisy, no longitudinal patient structure.
- **PubMed / PMC**: Open. Indexes broadly — vet science, traditional medicine, predatory journals, retracted papers mixed in.
- **MIMIC-CXR**: Radiology reports + chest X-rays. Silver labels from text.

Horses vs. zebras: Clinical notes are dominated by common conditions. PubMed **case reports** overrepresent rare findings — they invert the base-rate distribution. **Know your data source's selection bias before you train on it.**

► J Med Genet. 1977 Jun;14(3):190-193. doi: [10.1136/jmg.14.3.190](https://doi.org/10.1136/jmg.14.3.190) [cs](#)

Diabetes mellitus, diabetes insipidus, and optic atrophy. An autosomal recessive syndrome?

[F C Fraser, T Gunn](#)

► Copyright and License information

PMCID: PMC1013555 PMID: [881709](#)

Abstract

Twenty-one families were selected from the published reports in which the proband had the triad of juvenile diabetes mellitus, diabetes insipidus, and optic atrophy. The data were consistent with the hypothesis of an autosomal gene which, in the homozygote, causes juvenile diabetes mellitus and one or more of diabetes insipidus, optic atrophy, and nerve deafness. Heterozygotes appear to have an increased probability of developing juvenile diabetes mellitus.

A 1977 case report on a rare syndrome —
published *because* it's unusual.

How Text is Changing Right Now

Ambient AI Scribes: The Biggest Shift in Clinical Documentation

Ambient AI scribes (Nuance DAX Copilot, Abridge, Suki, etc.) listen to patient-provider conversations and auto-generate structured clinical notes.

What we know so far:

- Physicians report saving 1–3 hours/day on documentation.

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- The VA has deployed ambient scribes system-wide.
- Note quality is generally acceptable, with $\sim 1.5\%$ hallucination rate (Asgari et al., 2025).

Ambient AI Scribes: The Catch

The promise is “less documentation burden.” The reality is more nuanced:

- Notes become *longer and more detailed* — not shorter.

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Data type connection: The **generative origin** of clinical text is changing. Yesterday's training data was human-written under time pressure; tomorrow's is LLM-generated under revenue pressure. Same token vocabulary, different distribution.

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The pattern: Making documentation easier has increased text volume while *decreasing* information density at each step.

For ML researchers: The statistical properties of clinical text are a *moving target*.

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It's not just writing — LLMs are increasingly used to *process* clinical text:

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The implication: We are entering a world where LLMs both *write* and *read* clinical text. The human may be out of the loop in both directions. This raises fundamental questions about **factuality**, **accountability**, and **drift**.

Bringing It Together

Five Ways Clinical Text Breaks the “Sequence of Tokens” Abstraction

Text is a *sequence of nominal tokens* — but today we saw five ways this breaks down:

Issue	Where we saw it	ML consequence
Section structure	“MS” means different things in ROS vs. A&P; rigid rad report sections	BoW fails; need section-aware features
Temporal dependence	Copy-paste across visits; rad COMPARISON line	Split by patient, not note
Multi-modal links	Reports reference images, labs, orders	Text alone is incomplete
i.i.d. violation	50–80% duplication (heatmap)	Train–test leakage
Generative origin shift	AI scribes changing note statistics	Distribution drift

Bottom line: Clinical text *looks like* a standard NLP problem. The pathologies above are why it isn't.

Wrap-up

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Thursday: How do we *model* all of this? Domain adaptation, fairness in clinical NLP, factuality, and evidence-grounded generation.

References & Further Reading

Documentation burden & clinical text:

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ChatDoctor: huggingface.co/datasets/avaliev/chat.doctor

Questions?

Next: Lecture 20 — Clinical & Biomedical NLP: Modeling